Midterm 2 study guide

How to prepare — Lectures 18–32 and Quizzes 5–9

Last updated: October 8, 2026.

Exam: Midterm 2 · Friday, November 13 · in class. Practice problems: Practice packet. Schedule: Schedule.

Recommended study plan

1. **Scan the learning outcomes below.** Mark each as: confident / shaky / blank.
2. **Go through the lecture notes (18–32) and Labs 3–4.** For each outcome you marked shaky or blank, find the matching fill-in, worked example, or Check yourself item. (Skip Veterans Day — no new lecture.)
3. **Build your own "important definitions and formulas" list.** As you read, write down every definition or formula you might need on the exam (matrix sizes, Ax, RREF language, A^k, inverse tests, rank/null/col, determinants).
4. **Rework examples from lecture and quizzes.** Cover the solution, redo Warm-up / Worked / Check yourself / quiz-prep items by hand, then uncover and check using the filled-in notes in Canvas.
5. **Start memorizing your list of definitions and formulas.**
6. **Try the practice midterm problems with no notes.**
7. **Check answers** (end of the practice packet). For every miss, update your definitions/formulas list and redo a similar lecture example.
8. **Memorize the final list** so you can write it from memory before the exam. In class you will not have notes.

Useful Course materials

Source	Role
Lectures 18–32	Core ideas and skills (Unit II)
Quizzes 5–9	Fluency checkpoints (rework those styles; Quiz 5 is Unit I basis review)
Practice packet	Definitions, easy computations, harder items
Boyd & Vandenberghe, *VMLS*	Textbook (free online)
Strang §3.5	Rank, column space, null space (selected portions)

Learning outcomes (Lectures 18–32)

Light review — Quiz 5 / basis (Lecture 14)

1. Explain what a basis accomplishes and what dimension means.
2. Identify a basis and dimension in simple examples.
3. Use span, independence, basis, and dimension accurately when they reappear with matrices (column space, null space, rank).

Lecture 18 — Matrices as data and structured objects

1. Identify matrix dimensions, rows, columns, and entries.
2. Represent datasets, networks, images, or systems as matrices.
3. Perform matrix addition, scaling, and transposition.
4. Recognize zero, identity, diagonal, triangular, symmetric, and sparse matrices.
5. Explain one practical consequence of matrix structure.

Lecture 19 — Matrix-vector multiplication

1. Compute Ax by hand for a small matrix.
2. Interpret Ax as a linear combination of the columns of A.
3. Interpret A as an input-output transformation.
4. Explain how the dimensions of A determine the input and output spaces.

Lecture 20 — Transforming space (+ Lab 3)

1. Predict the effect of scaling, reflection, shear, and rotation matrices.
2. Apply matrices to many 2D or 3D points at once.
3. Plot and compare original and transformed objects.
4. Adapt a short program that implements a transformation.
5. Interpret a matrix as a coordinate or physical-space transformation.

Lecture 21 — Systems and $Ax = b$

1. Translate equations or contextual constraints into $Ax = b$.
2. Interpret $Ax = b$ as asking whether b belongs to the span of the columns of A.
3. Classify systems as square, underdetermined, or overdetermined.
4. Anticipate circumstances producing zero, one, or many solutions.

Lecture 22 — Gaussian elimination

1. Perform elementary row operations.
2. Reduce a small augmented matrix to echelon form.
3. Solve a small system using back substitution.
4. Explain why row operations preserve the solution set.

Lecture 23 — RREF, pivots, free variables, and software

1. Interpret reduced row echelon form.
2. Identify pivot and free variables.
3. Describe a solution set parametrically.
4. Distinguish inconsistent, unique, and infinite-solution cases.
5. Use software to solve or row-reduce a system and verify with a residual.

Lecture 24 — Linear dynamical systems

1. Represent an evolving system as $x_{t+1} = Ax_t$.
2. Interpret the state vector and transition matrix.
3. Build a transition matrix from a flow description.
4. Compute and interpret the next few states.

Lecture 25 — Simulation and long-run behavior

1. Simulate many time steps computationally.
2. Plot state components through time.
3. Recognize stabilization, growth, decay, or oscillation.
4. Conduct a simple what-if simulation.
5. State limitations of a linear dynamical model.

Lecture 26 — Predict a system through time (+ Lab 4)

1. Implement a linear dynamical model.
2. Write or complete a short iterative calculation.
3. Validate the first steps by hand.
4. Compare results under altered parameters or initial states.
5. Record an unexplained long-run pattern for later investigation (eigenvalues come later).

Lecture 27 — Matrix multiplication

1. Determine whether a product is defined and give its dimensions.
2. Compute a small matrix product.
3. Interpret ABx as a sequence of transformations.
4. Explain why matrix multiplication is generally not commutative.

Lecture 28 — Composition and matrix powers

1. Combine multistage processes using a matrix product.
2. Connect $A^k x$ to repeated dynamics.
3. Interpret selected entries of a matrix product or power.
4. Recognize how network and transition information accumulates through powers.

Lecture 29 — Computational cost

1. Compare algebraically equivalent orders of computation.
2. Count intermediate values that must be stored.
3. Measure runtime for increasing problem sizes.
4. Explain why forming a large intermediate matrix may be wasteful.
5. Choose a computational order and justify it quantitatively.

Lecture 30 — Inverses and solving

1. Interpret an inverse as undoing a transformation.
2. Determine whether a simple matrix is invertible.
3. Connect invertibility to uniqueness of $Ax = b$.
4. Explain why solving $Ax = b$ is generally preferable computationally to explicitly calculating A^{-1}.

Lecture 31 — Rank, column space, null space

1. Interpret column space and null space through $Ax = b$ and $Ax = 0$.
2. Determine rank from pivots or independent columns.
3. Find simple bases for column or null spaces.
4. Connect rank to consistency, independence, and dimension.
5. Use subspace, basis, dimension, rank, column space, and null space correctly.

Lecture 32 — Determinants, geometry, and invertibility

1. Compute determinants of 2×2 and simple triangular matrices.
2. Interpret determinant magnitude as area or volume scaling.
3. Connect determinant zero with lost dimension and noninvertibility.
4. Recognize the determinant of a triangular matrix from its diagonal.
5. Connect determinants to the coming characteristic equation (preview only).
